

Naïve Bayes

Bayes Theorem

Bayes' theorem helps you examine the probability of an event based on the prior knowledge of any event that has correspondence to the former event. Its uses are mainly found in probability theory and statistics. The formula for Bayes' theorem is given as:

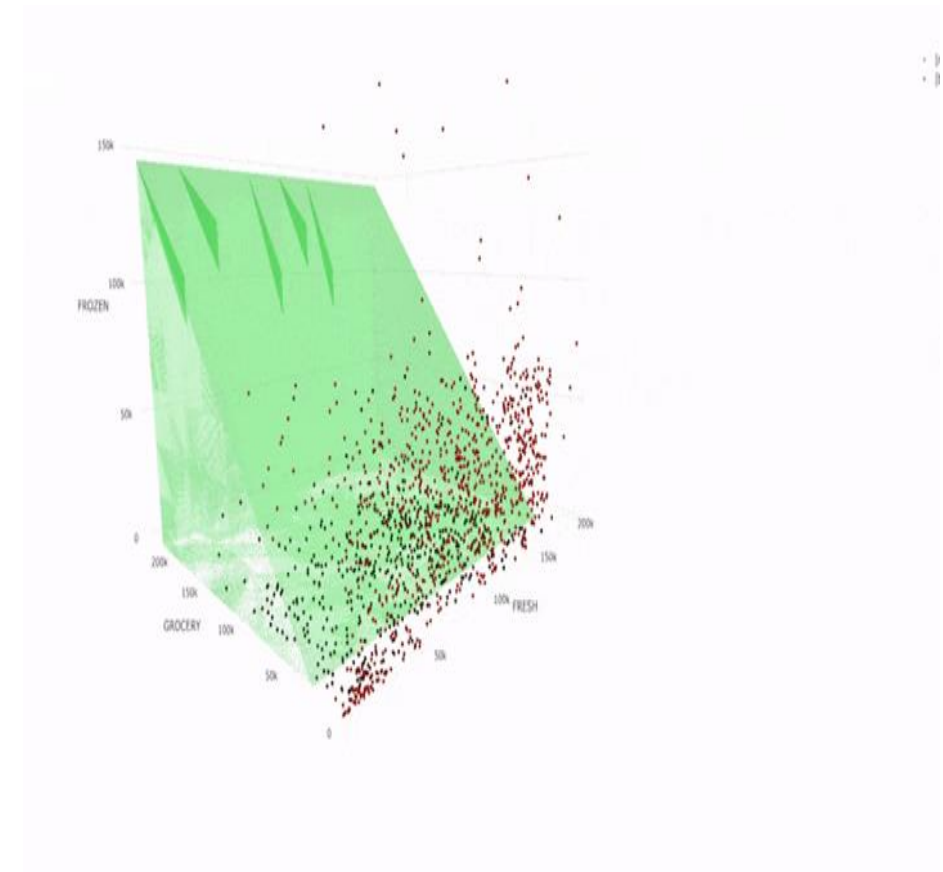
$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Where, A and B are called events.

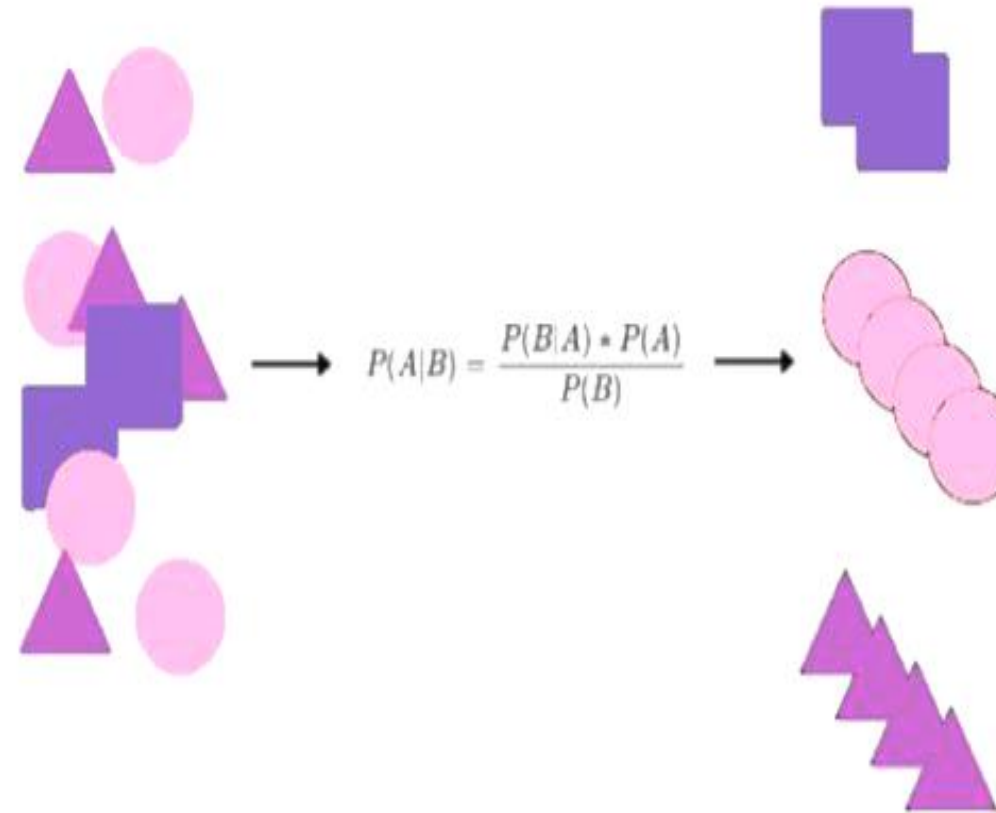
- $P(A | B)$ is the probability of event A, given the event B is true (has occurred). Event B is also termed as evidence.
- $P(A)$ is the priori of A (the prior independent probability, i.e. probability of event before evidence is seen).
- $P(B)$ is the priori of B (the prior independent probability, i.e. probability of event before evidence is seen).
- $P(B | A)$ is the probability of B given event A, i.e. probability of event B after evidence A is seen.

What Is Naive Bayes?

- Naive Bayes is a supervised learning algorithm, based on Bayes theorem and used to solve classification problems.
- It can be easily written in code and predictions can be made real quick, which in turn increases the scalability of the solution.
- It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.



- Naive Bayes Classifier is one of the simple and **most effective classification algorithms** which helps in building the fast machine learning models that can make **quick predictions**.
- Some popular examples of Naive Bayes Algorithm are **spam filtration**, **Sentimental analysis**, and **classifying articles**.



Why is it called Naive Bayes?

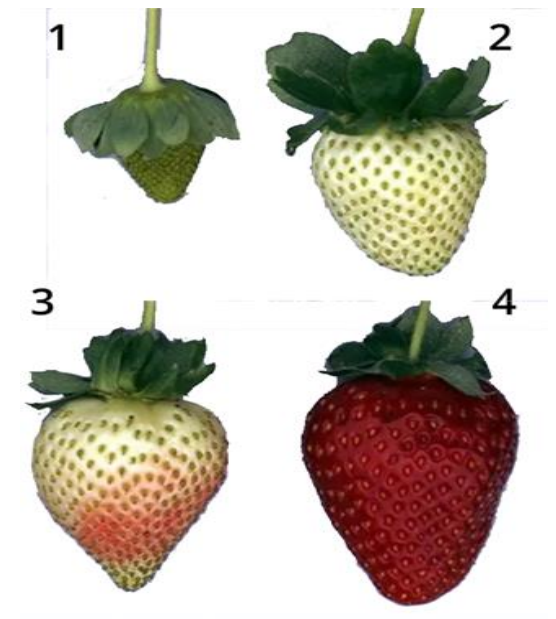
Naive Bayes algorithm is comprised of two words **Naive** and **Bayes**, which can be Summarized as:

Naive — It is called Naive because it assumes that the occurrence of a certain feature is independent of the occurrence of other features. Which might not be true in real life.

Example: **If a strawberry is ripe or not to harvest?**

Assume, two independent features in this case: **Size** and **Color**. To demonstrate the assumptions are partially invalid, pose the following two critical questions:

- Size and Color are independent? Not really, a positive correlation could be seen apparently. The fruit grows, expanding in size and changing in color.
- Size and Color equally contribute to the outcome of “Being ripe”? Not really! Although it greatly depends on the type of fruit, we can roughly anticipate the answer.



Bayes — It is called Bayes because it depends on the principle of Bayes' Theorem.

How do Naive Bayes Algorithm work?

- Convert the data set into a frequency table.
- Create Likelihood table by finding the probabilities.
- Use Naive Bayesian equation to calculate the posterior probability.

Weather	Play
Sunny	No
Overcast	Yes
Rainy	Yes
Sunny	Yes
Sunny	Yes
Overcast	Yes
Rainy	No
Rainy	No
Sunny	Yes
Rainy	Yes
Sunny	No
Overcast	Yes
Overcast	Yes
Rainy	No

Frequency Table		
Weather	No	Yes
Overcast		4
Rainy	3	2
Sunny	2	3
Grand Total	5	9

Likelihood table				
Weather	No	Yes		
Overcast		4	=4/14	0.29
Rainy	3	2	=5/14	0.36
Sunny	2	3	=5/14	0.36
All	5	9		
	=5/14	=9/14		
	0.36	0.64		

Types of Naive Bayes Classifier

- **Gaussian Naive Bayes** — In a Gaussian Naive Bayes, the predictors take a continuous value assuming that it has been sampled from a Gaussian Distribution. It is also called a Normal Distribution.
- **Multinomial Naive Bayes** — These types of classifiers are usually used for the problems of document classification. It checks whether the document belongs to a particular category like sports or technology or political etc and then classifies them accordingly. The predictors used for classification in this technique are the frequency of words present in the document.
- **Bernoulli Naive Bayes** — This classifier is also analogous to multinomial naive bayes but instead of words, the predictors are Boolean values. The parameters used to predict the class variable accepts only yes or no values, for example, if a word occurs in the text or not.

Pros & Cons of Naive Bayes

Pros of Naive Bayes —

- It is **easy** and **fast** to predict the class of the training data set.
- It performs well in **multiclass prediction**.
- It performs better as compared to other models like **logistic regression** while assuming the independent variables.
- It requires less training data.
- It performs better in the case of **categorical input** variables as compared to **numerical variables**.

Pros & Cons of Naive Bayes

Cons of Naive Bayes-

- The model will assign a 0 (zero) probability and will be unable to make a prediction if categorical variable has a category (in test data set). This is often known as “Zero Frequency”. To solve this, we can use the smoothing technique like “Laplace estimation”.
- Another limitation of this algorithm is the assumption of independent predictors. In real life, it is almost impossible that we get a set of predictors which are completely independent.
- Naive Bayes is also known as a **bad estimator**, so the probability outputs are not taken seriously.

Problem: Understanding Naive Bayes

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

(Outlook = sunny, Temperature = cool, Humidity = high, Wind = strong)

Check probability for playing tennis, when Outlook=Sunny, Temperature=Hot and Wind=Strong.

$$P(\text{PlayTennis} = \text{yes}) = 9/14 = .64$$

$$P(\text{PlayTennis} = \text{no}) = 5/14 = .36$$

Outlook	Y	N		Humidity	Y	N
sunny	2/9	3/5		high	3/9	4/5
overcast	4/9	0		normal	6/9	1/5
rain	3/9	2/5				
Temperature				Windy		
hot	2/9	2/5		Strong	3/9	3/5
mild	4/9	2/5		Weak	6/9	2/5
cool	3/9	1/5				

Understanding Naive Bayes

Shopping Example

Based on this dataset containing three types of input Day, Discount and Free Delivery. We will populate frequency tables for each attribute.

1	Day	Discount	Free Delivery	Purchase
2	Weekday	Yes	Yes	Yes
3	Weekday	Yes	Yes	Yes
4	Weekday	No	No	No
5	Holiday	Yes	Yes	Yes
6	Weekend	Yes	Yes	Yes
7	Holiday	No	No	Yes
8	Weekend	Yes	No	Yes
9	Weekday	Yes	Yes	Yes
10	Weekend	Yes	Yes	Yes
11	Holiday	Yes	Yes	Yes
12	Holiday	No	Yes	Yes
13	Holiday	No	No	No
14	Weekend	Yes	Yes	Yes
15	Holiday	Yes	Yes	Yes
16	weekday	Yes	Yes	Yes
17	weekday	Yes	Yes	Yes
18	weekend	No	Yes	No
19	weekday	Yes	No	Yes
20	weekday	No	Yes	Yes

21	weekday	No	No	No
22	weekday	Yes	Yes	Yes
23	weekday	Yes	Yes	Yes
24	weekend	Yes	Yes	Yes
25	weekend	Yes	Yes	Yes
26	weekend	Yes	Yes	Yes
27	Holiday	Yes	Yes	Yes
28	Holiday	No	Yes	Yes
29	Holiday	No	Yes	Yes
30	Holiday	No	Yes	No
31	Holiday	Yes	No	No

For our Bayes Theorem, Let the event Buy be **A** and the **independent** variable **Discount, Free Delivery and Day** be **B**. Let's look into the representation of the events in the below figure.

The diagram illustrates three frequency tables for variables related to the event Buy (A). Each table is a 2x2 contingency table with 'Buy' as the column header and the variable as the row header. The variables are Discount, Free Delivery, and Day. The 'Buy' column is split into 'Yes' and 'No' categories. The 'Discount' and 'Free Delivery' tables have 'Yes' and 'No' as row categories. The 'Day' table has 'Weekday', 'Weekend', and 'Holiday' as row categories. Dashed arrows indicate the relationship between these variables and the event A, and a combined dashed arrow points to B.

Frequency Table		Buy	
		Yes	No
Discount	Yes	19	1
	No	5	5

Frequency Table		Buy	
		Yes	No
Free Delivery	Yes	21	2
	No	3	4

Day	Buy	
	Yes	No
	Weekday	9
Weekend	7	1
Holiday	8	3

Likelihood Table for the variable
Day, Discount and Free Delivery
with probabilities:

Likelihood Tables

Likelihood Table		Buy		
		Yes	No	
Day	Weekday	9/24	2/6	11/30
	Weekend	7/24	1/6	8/30
	Holiday	8/24	3/6	11/30
		24/30	6/30	

Frequency Table		Buy		
		Yes	No	
Discount	Yes	19/24	1/6	20/30
	No	5/24	5/6	10/30
		24/30	6/30	

Frequency Table		Buy		
		Yes	No	
Free Delivery	Yes	21/24	2/6	23/30
	No	3/24	4/6	7/30
		24/30	6/30	

Based on the table Day conditional probabilities are given below,

$$P(\mathbf{B}) = P(\mathbf{Weekday}) = 11/30 = 0.367$$

$$P(\mathbf{A}) = P(\mathbf{Buy}) = 24/30 = 0.8, P(\mathbf{A}) = P(\mathbf{No Buy}) = 6/30 = 0.2$$

$$P(\mathbf{B} | \mathbf{A}) = P(\mathbf{Weekday} | \mathbf{Buy}) = 9/24 = 0.375, P(\mathbf{B} | \mathbf{A}) = P(\mathbf{Weekday} | \mathbf{No Buy}) = 2/6 = 0.33$$

If **A** equals **Buy** then:

$$P(\mathbf{A} | \mathbf{B}) = P(\mathbf{Buy} | \mathbf{Weekday}) = P(\mathbf{Weekday} | \mathbf{Buy}) * P(\mathbf{Buy}) / P(\mathbf{Weekday})$$

$$= (0.375 * 0.8) / 0.367 = 0.817$$

Again,

If **A** equals **No Buy** then:

$$P(\mathbf{A} | \mathbf{B}) = P(\mathbf{No Buy} | \mathbf{Weekday}) = P(\mathbf{Weekday} | \mathbf{No Buy}) * P(\mathbf{No Buy}) / P(\mathbf{Weekday})$$

$$= (0.33 * 0.2) / 0.367 = 0.179$$

As the Probability(**Buy | Weekday**) is **more** than the Probability(**No Buy | Weekday**) , we can conclude that a customer will most likely buy the product on a Weekday.

Now we will use the three likelihood tables to calculate whether a customer will purchase a product on a specific combination of Day=Holiday with Free Delivery and Discount.

Calculating Conditional Probability of purchase on the following combination of day, discount and free delivery:

Where B equals:

- Day = Holiday
- Discount = Yes
- Free Delivery = Yes

Let A = No Buy

$P(A|B) = P(\text{No Buy} \mid \text{Discount} = \text{Yes}, \text{Free Delivery} = \text{Yes}, \text{Day} = \text{Holiday})$

$$= \frac{P(\text{Discount} = \text{Yes} \mid \text{No}) * P(\text{Free Delivery} = \text{Yes} \mid \text{No}) * P(\text{Day} = \text{Holiday} \mid \text{No}) * P(\text{No Buy})}{P(\text{Discount} = \text{Yes}) * P(\text{Free Delivery} = \text{Yes}) * P(\text{Day} = \text{Holiday})}$$

$$= \frac{(1/6) * (2/6) * (3/6) * (6/30)}{(20/30) * (23/30) * (11/30)}$$

$$= 0.178$$

Calculating Conditional Probability of purchase on the following combination of day, discount and free delivery:

Where B equals:

- Day = Holiday
- Discount = Yes
- Free Delivery = Yes

Let A = Buy

$P(A|B) = P(\text{Yes Buy} \mid \text{Discount} = \text{Yes}, \text{Free Delivery} = \text{Yes}, \text{Day} = \text{Holiday})$

$$= \frac{P(\text{Discount} = \text{Yes} \mid \text{Yes}) * P(\text{Free Delivery} = \text{Yes} \mid \text{Yes}) * P(\text{Day} = \text{Holiday} \mid \text{Yes}) * P(\text{Yes Buy})}{P(\text{Discount} = \text{Yes}) * P(\text{Free Delivery} = \text{Yes}) * P(\text{Day} = \text{Holiday})}$$

$$= \frac{(19/24) * (21/24) * (8/24) * (24/30)}{(20/30) * (23/30) * (11/30)}$$

$$= 0.986$$

Probability of **purchase** = 0.986

Probability of **no purchase** = 0.178

Now normalizing these conditional probabilities to get the likelihood of this events.

Sum of the probabilities = $0.986 + 0.178 = 1.164$

Likelihood of purchase = $0.986/1.164 = 84.71\%$

Likelihood of no purchase = $0.178/1.164 = 15.29\%$

As **84.71%** is **greater** than **15.29%**, we can conclude that an average customer will buy on a Holiday with Discount and Free Delivery.

When to use Naive Bayes

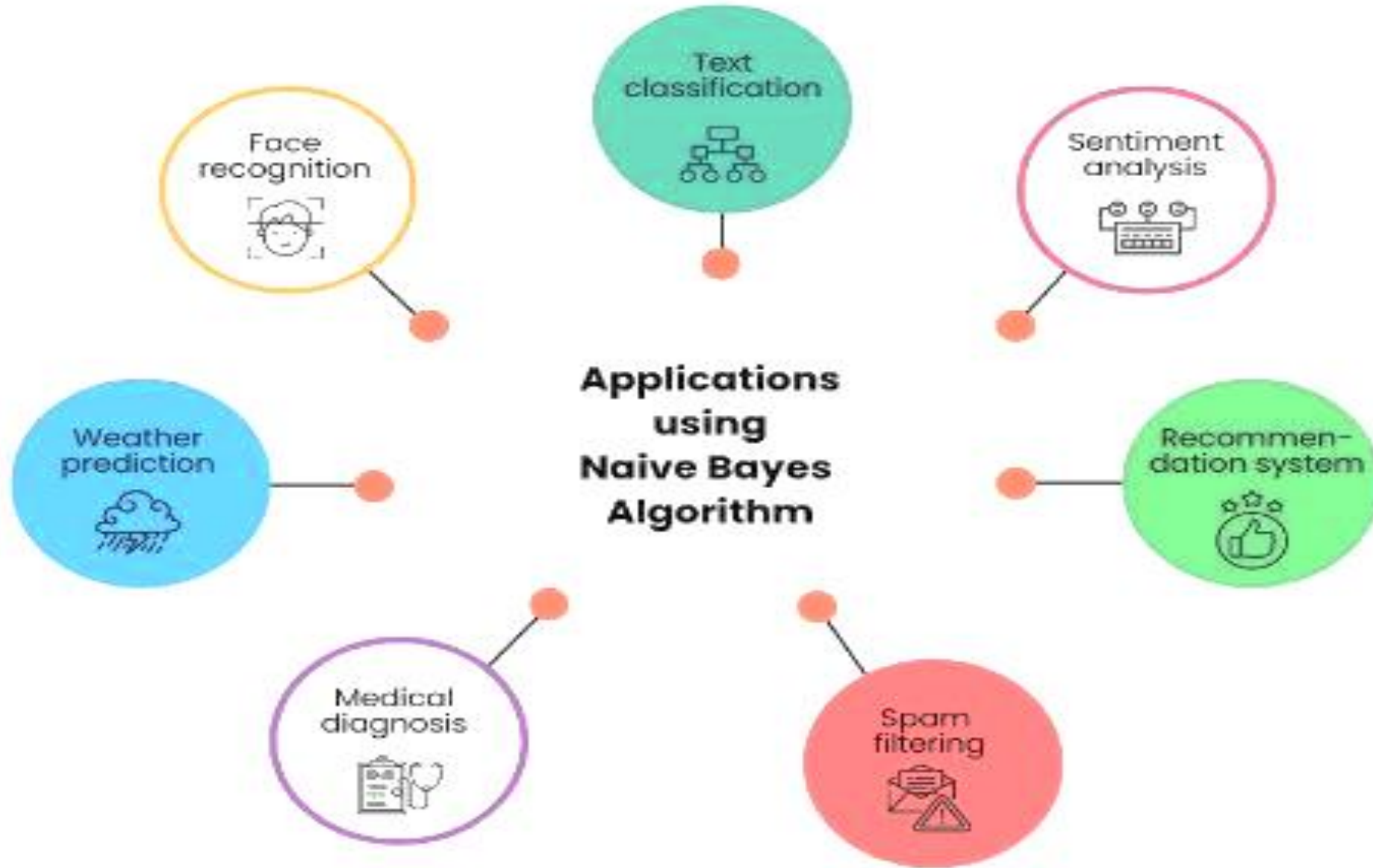
A Naive Bayesian classifier performs worse than a complex classifier due to the strict assumptions it makes about the data. The classifier, however, has some patterns:

- Training and predicting the model is done at a high speed.
- Probabilistic predictions can be created purely based on the data.
- They are usually pretty easy to interpret.
- Their parameters are usually not tunable.

When to Use Naive Bayes

- 1 High-speed training and prediction
- 2 Probability prediction based on data
- 3 Easy to interpret
- 4 Non-tunable parameters

Application of Naive Bayes Classifier



Application of Naive Bayes Classifier

In the field of machine learning, Naive Bayes is used as a classification model, i.e. to classify a data set into a certain class. There are various concrete applications for these models for which Naive Bayes is also used:

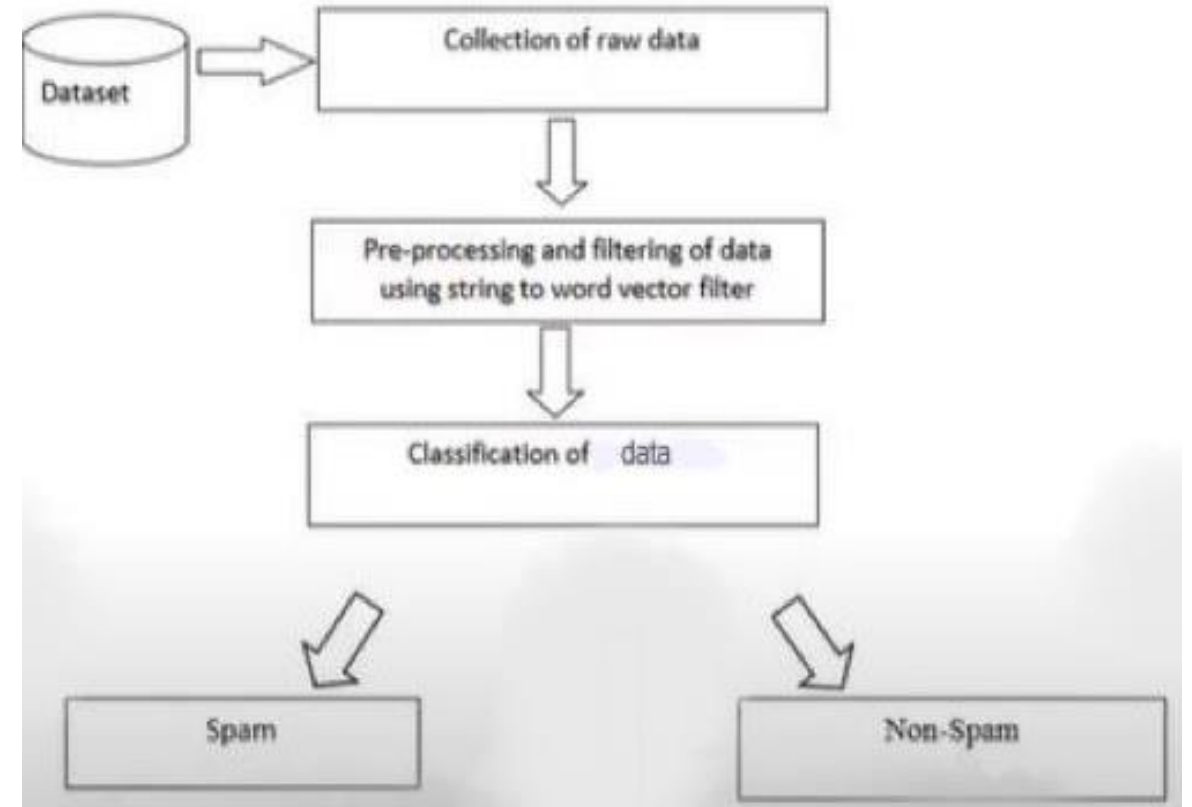
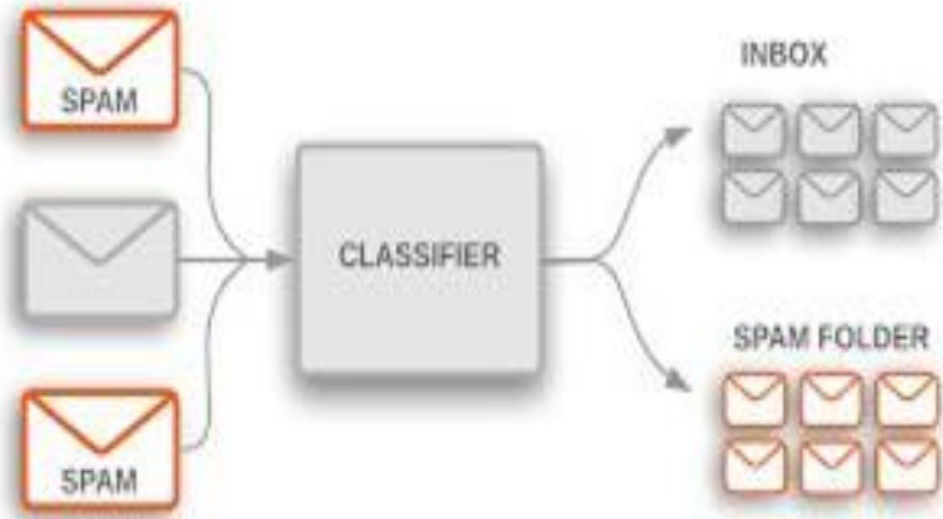
- Real-time prediction.
- Multi-class prediction.
- Recommendation Systems.
- Classification of credit risk.
- Face Recognition.
- Predict Medical treatment.
- Text Classification/ Spam Filtering / Sentiment Analysis :

They are mostly used in text classification problems because of its multi-class problems and the independence rule. They are used for identifying spam emails and also to identify negative and positive customer sentiments on social platforms.

Application of Naive Bayes Classifier

Spam Classification

Flowchart



1. Diagram for classifying Spam text data.

Application of Naive Bayes Classifier

Spam Classification

Suppose we have a training set that contains the following:

	comment	category
0	Check out my you[tube] /#?song Channel?	spam
1	I love song, please like my channel!!!!	spam
2	I'm not a fan, :(	ham
3	This song is great, I love your channel	ham
4	love love love	ham

Suppose we have a new comment:

I love song

Would this be categorized as:

Spam/Ham?

2. Training set with instances, if the comments are spam or not. These will be used for the new comment.

Application of Naive Bayes Classifier

Spam Classification

Extract Features:



3.Feature extraction

Application of Naive Bayes Classifier

Spam Classification

	channel	check	fan	great	like	love	song	tube	class
0	1	1	0	0	0	0	1	1	spam
1	1	0	0	0	1	1	1	0	spam
2	0	0	1	0	0	0	0	0	ham
3	1	0	0	1	0	1	1	0	ham
4	0	0	0	0	0	3	0	0	ham

positions ← all word positions in test document

$$c_{NB} = \underset{c \in C}{\operatorname{argmax}} P(c) \prod_{i \in \text{positions}} P(w_i | c)$$

$$P(c=\text{spam} | \text{I love song}) = P(\text{Spam}) \times P(\text{I love song} | \text{Spam})$$

$$P(\text{Ham} | \text{I love song}) = P(\text{Ham}) \times P(\text{I love song} | \text{Ham})$$

First, we calculate the apriori probability that the document is spam and the document is ham, which can easily be obtained by counting the number of documents in each class dividing it by the total number of documents.

	channel	check	fan	great	like	love	song	tube	class
0	1	1	0	0	0	0	1	1	spam
1	1	0	0	0	1	1	1	0	spam
2	0	0	1	0	0	0	0	0	ham
3	1	0	0	1	0	1	1	0	ham
4	0	0	0	0	0	3	0	0	ham

positions ← all word positions in test document

$$c_{NB} = \underset{c \in C}{\operatorname{argmax}} P(c) \prod_{i \in \text{positions}} P(w_i | c)$$

$$P(c = \text{ham}) = \frac{N_{\text{ham}}}{N_{\text{doc}}} \quad P(c = \text{spam}) = \frac{N_{\text{spam}}}{N_{\text{doc}}}$$

$$P(\text{spam}) = \frac{\# \text{ of Spam comments}}{\# \text{ total comments}} = \frac{2}{5}$$

$$P(\text{ham}) = \frac{\# \text{ of Ham comments}}{\# \text{ total comments}} = \frac{3}{5}$$



If not spam, then the email is ham. Can't be both.
The probability of spam and ham will sum up to 1.
 $P(\text{ham}) = 1 - P(\text{spam})$. Remember your stat 105?

4. Finding if the text will be Spam or Ham.

Application of Naive Bayes Classifier

Spam Classification

	channel	check	fan	great	like	love	song	tube	class
0	1	1	0	0	0	0	1	1	spam
1	1	0	0	0	1	1	1	0	spam
2	0	0	1	0	0	0	0	0	ham
3	1	0	0	1	0	1	1	0	ham
4	0	0	0	0	0	3	0	0	ham

positions $\leftarrow$ all word positions in test document

$$c_{NB} = \underset{c \in C}{\operatorname{argmax}} P(c) \prod_{i \in \text{positions}} P(w_i|c)$$

$$P(w_i|\text{spam}) = \frac{\text{count}(w_i, \text{spam}) + \alpha}{(\sum_{w \in V} \text{count}(w, \text{spam})) + \alpha|V|}$$

$$P(\text{love}|\text{spam}) = \frac{1+1}{8+8} = \frac{2}{16}$$

$$P(\text{song}|\text{spam}) = \frac{2+1}{8+8} = \frac{3}{16}$$

$$P(w_i|\text{ham}) = \frac{\text{count}(w_i, \text{ham}) + \alpha}{(\sum_{w \in V} \text{count}(w, \text{ham})) + \alpha|V|}$$

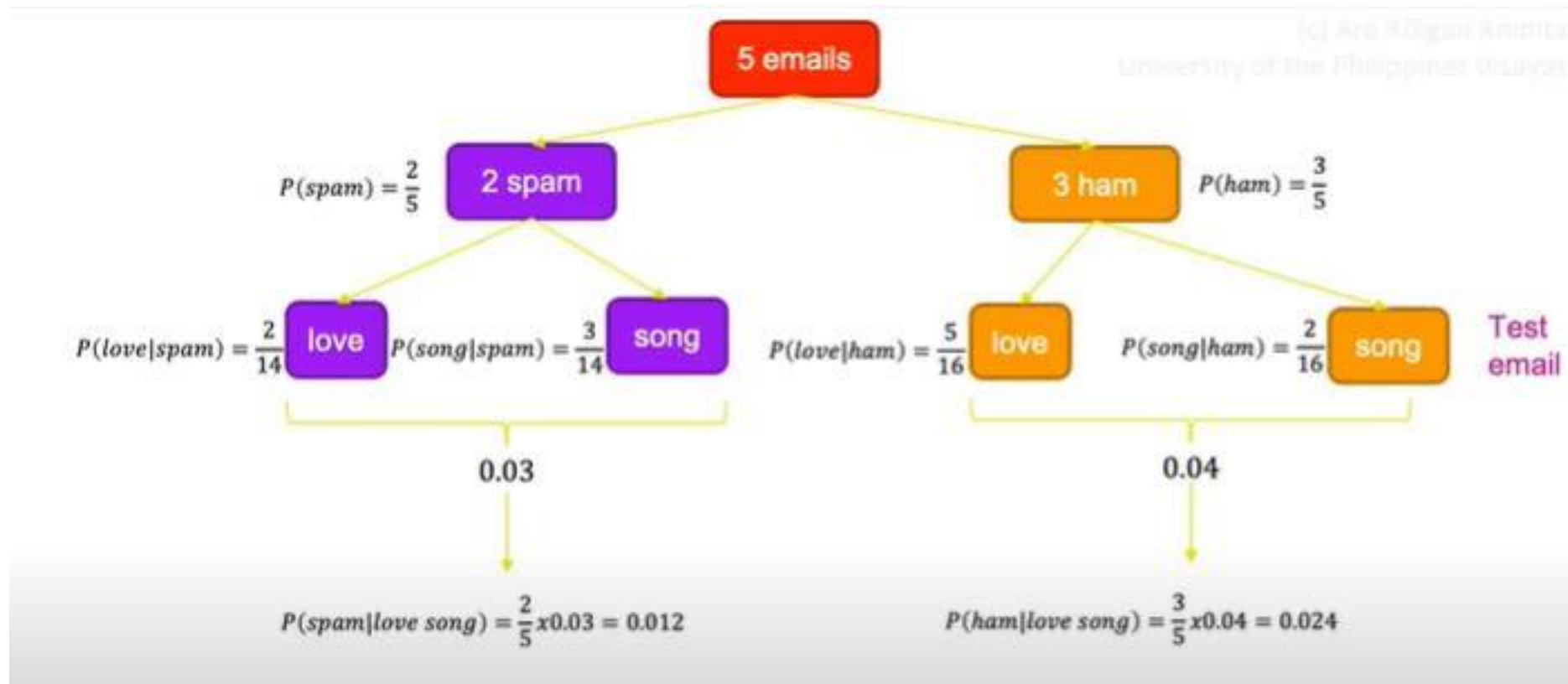
$$P(\text{love}|\text{ham}) = \frac{4+1}{8+8} = \frac{5}{16}$$

$$P(\text{song}|\text{ham}) = \frac{1+1}{8+8} = \frac{2}{16}$$

4. Finding if the text will be Spam or Ham (cont.).

Application of Naive Bayes Classifier

Spam Classification



5. Comparing the values, we found that the text “I love song” is **not spam or ham** as we got **greater** value after calculation.

THANK YOU !!